



Analytical and experimental investigation of steel friction dampers and horizontal brake pads in chevron frames under cyclic loads

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ABSTRACT

Recent observations of earthquake-related damage have shown that some structural damage occurs in welded joints. Because of weak connections, the idea of using chevron bracing with a horizontal friction brake pad (FBP) damper, which can easily be replaced after an earthquake, has been proposed. First, in this study, a high-performance friction damper that can be manufactured and installed at a low cost is introduced. First, nine models were analysed numerically with ABAQUS software, and then two optimal models were selected and tested in a structural laboratory. ABAQUS software was also used to model the optimal damper inside a chevron frame and analyse its performance. In addition, three 8-story buildings were considered, one with a moment-resisting frame (MRF), one with a structural bracing system with an added damping and stiffness (ADAS) damper, and one with an FBP damper. Then, eleven far-field earthquake records were used for nonlinear time history analysis. Next, the effect of the friction damper between two adjacent buildings was investigated by modelling a 5-story structure using OpenSees software under seven earthquake records. The results show that a very high stress concentration occurs in the damper area of the brake pad after a load is applied to the bracing frame, while dampers in other elements, such as beams and columns, provide less stress. The ten-screw model has the highest energy absorption, according to parametric studies. Compared to other frames, the chevron frame with a friction damper exhibits a 32% increase in energy absorption. By analysing other seismic parameters of the frame, the damper improves frame performance

1. Introduction

Earthquake losses, in financial terms and human lives, highlight the need for structural engineers to understand earthquake behaviour. Moreover, choosing a lateral bracing system that is both high-performing and affordable can be a challenging task for engineers. One of the most common methods for resisting earthquake forces is by using convergent braces in the chevron frame. Structural design considers the strength, ductility, and energy absorption capacity of a potential structure. A significant challenge in achieving this goal is applying and presenting new methods in the seismic design of structures to improve consumable quality. In this context, structural design engineers have focused their attention on new methods to address this

phenomenon, such as the use of friction dampers, which can control vibrations in structures. A damping system that joins a structure can increase its energy dissipation capacity and reduce or minimize structural damage. This study investigates friction dampers as an energy dissipation tool. As the number of vibration control methods of structures has increased, mainly passive methods because they consume seismic energy, the need to produce and apply such devices in the construction industry has grown. One of the primary ways to stop moving objects is to brake. Dampers can be produced and installed at a low cost because of their simple form. Dry friction dampers are used to increase damping in structures. These dampers are installed at the intersection of the chevron brace and the main beam and consume energy to convert kinetic energy into heat during an earthquake. Because

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the major components of the structure remain the same, we can also use these devices in the reconstruction or strengthening of the structure. The friction damper helps the building respond to vibration and dissipate seismic energy. It can optimize structures to reduce costs, which leads to significant savings. This energy-dissipating device uses a friction joint with long slot holes; if a force exceeds the slip load on the joint, then the surfaces will suddenly begin to move. Based on the materials used for the contact surfaces, the frictional force changes after movement. Such a damper is used if it can create a frictional force in a joint or member to control the amount of force.

Materials used for the slip surfaces include brake pads on metal, metal on brass, and metal on metal. The choice of base metal is essential for friction dampers. High corrosion resistance can often reduce the assumed coefficient of friction for the desired life of a device. The reason for choosing friction dampers is the simplicity of the manufacturing mechanism, the lack of special and expensive materials, and finally, the ability to manufacture and test these devices. These advantages of the efficiency of friction dampers make them a suitable option to begin investigating structural control methods. The friction of two solid bodies in contact has an essential role in controlling displacement and energy loss. For example, friction in car brakes can also waste kinetic energy on a much smaller scale. In addition to their simple shape, friction dampers can be produced and installed at a low cost. Furthermore, the friction force can be controlled by adjusting the prestressing force of the screws. The seismic design theory for frame structures equipped with friction dampers was developed by Pall and Marsh [1]. Passive control devices have been proposed to dissipate input energy in structures [2]. Pall proposed a limited-slip bolted (LSB) joint [1]. Another type of friction damper, a slotted bolted connection (SBC), was developed at San Jose State University. These results encouraged more researchers to evaluate the performance of this type of damper [3]. Fitzgerald and Anagnos presented more comprehensive results about friction dampers in seismic response control [4]. Constantinou et al. introduced friction dampers composed of two friction discs assembled and fastened with adjustable screws [5].

Because of its mechanical properties, brass has increased abrasion resistance and improved the performance of friction devices [6]. The seismic performance of a structural model using a friction damper was studied by Reinhorn and Li [7]. The rotary friction damper was developed by Mualla [8]. A new friction damper based on rotational motion was implemented by Mualla and Belev on chevron braces [9]. Lu proposed a semiactive control method for determining friction-controlled clamping forces [10]. A study of the seismic response of two adjacent buildings connected to friction dampers under different earthquakes was conducted by Bhaskararao and Jangid [11]. A formula for the equations of motion for two adjacent buildings with multiple degrees of freedom (MDOF) is provided with associated friction dampers. A new algorithm was developed to simulate the seismic response of an N-story building frames equipped with energy-dissipating devices, which also modifies the linear acceleration method [12]. An earthquake design method was proposed based on the shear force distribution of elastic structures using friction dampers [13]. The rotary friction damper designed by Gholizad and Golafshani has been used to strengthen a jacket [14]. The behaviour of an eccentric bracing frame equipped with a friction damper at the junction of cables was examined [15]. Ozbulut and Hurlbeaus proposed two fuzzy logic controller methods to control the piezoelectric force of friction dampers installed at the base of structures to prevent damage to the structures against various types of seismic excitations [16].

Kim and Choi studied the increase in seismic resistance and the exponential collapse of structures with friction dampers [17]. To identify the mechanical properties of the brake pads, studies have been performed by Orłowicz A et al. The friction coefficient of disc brake pads was experimentally investigated in their work. Nine minutes from the beginning of an experiment, the results showed that it reaches its optimal performance at $\mu = 0.41$, which then decreases and stabilizes at $\mu = 0.38$. Brake pads can be described as friction plates with suitable

properties [18]. Using the capacity spectrum method, these authors designed friction dampers and tested them for seismic and collapse resistance. Experimental research was conducted on the eccentrically braced frame (EBF) at a structural laboratory after installing dampers at the frame's intersection with the braces [19]. Golondrino et al. also developed asymmetrical friction connections (AFCs). Additionally, the brake pads' effective friction coefficient was tested and studied [20]. Naeem and Kim developed a new device to induce displacement-dependent energy to protect structures from earthquakes [21]. Sano et al. introduced a multiunit friction damper using high-tension bolts tightened via coned disc springs [22]. An experimental study on a friction damper with shear plates, studs, and disc nut bolts was presented by Sano and Shirai [23]. Wang et al. presented a comprehensive study of self-propelled dampers equipped with friction springs. That study examined how the taper surfaces of the friction springs affected the overall damper behaviour in three dampers. In the damper specimens, the friction ring springs showed the expected deformation, while the other components remained lightly stressed [24]. A variable-friction self-centring energy-dissipation brace (VF-SCEDB) was introduced by Chen et al. [25]. In that research, three samples of dampers were tested in the laboratory, and the effect of the operation of the conical surfaces of friction springs on the overall behaviour of the damper was investigated. According to the results, the VFD provided a more stable dissipation rate and higher self-centering tendency than the flat friction device.

Wang et al. proposed a dual self-centring variable friction damper (DSC-VFD) brace with high postyield stiffness and a high energy dissipation potential [26]. Sui et al. proposed an optimal friction damper with friction pads for stable energy dissipation. That experiment used disc friction pads in a smooth and sandblasted apparatus [27]. Golmoghany and Zahrai introduced a friction damper with a vertical shear panel to improve the seismic performance of structures. By reducing the drift of the frame, the addition of this damper resulted in improved performance [28]. Fibres and organic and inorganic fillers were used in friction pads developed by Paronesso and Lignos. The resulting performance was fairly invariant regarding temperature and sliding velocity, as long as the velocity was faster than 10 mm/s [29]. A knee brace frame with a friction damper was studied numerically by Ghafouri-Nejad and Alirezaei et al. [30].

Observations of damage to buildings during recent earthquakes have shown that some of the damage was in the area of welded connections. The main purpose of this study is to introduce the *FBP* damper as an efficient damper to reduce damage in welded beam-to-column joints and increase energy absorption in chevron frames. This study also investigates the proper performance of this damper in medium-rise structures and addresses adjacent buildings.

This paper presents a type of friction damper that is resistant to cyclic loads and thus is stable. This damper consumes a high percentage of energy, preventing the destruction and damage of structural load-bearing elements. This new device exhibits a consistent dynamic behaviour during loading, and even in multiple loading cycles, no change in its performance has been observed. Laboratory studies also confirm the optimal performance of this damper in dynamic loads. This damper is studied because it can be built and installed at a low cost and high efficiency and later replaced after an earthquake. The difference between the current research and other previous studies on friction dampers is that the proposed *FBP* damper consists of two brake pad plates combined with steel plates and has a unique shape, given the pattern of the arrangement of the plates and screws; additionally, the installation location of this damper in a chevron frame with a horizontal function is considered to be innovative. Drift is required to activate these dampers because they act in shear. Therefore, they should be installed in places with sufficient drift. One of the most suitable places for installing friction dampers is the location of the intersection of the braces, which are activated by the in-story drift. In this research, an attempt has been made to place the damper in the frame so that it does not deviate from

the axis. It has been observed by numerical and laboratory study that the behaviour of this proposed type of connection is appropriate and exhibits good performance in absorbing lateral forces and earthquake motions. Finally, the innovation of this research is a damper in the proposed geometry and shape, and in the second stage of innovation, its performance involves the frame. A characteristic of this research is that brake pads have been used. Due to its mechanical properties, such as a high friction coefficient, the bracing frame equipped with the *FBP* damper is more efficient and absorbs more energy; additionally, the geometric design of the *FBP* damper helps to optimize and increase the efficiency of this passive control system.

2. Theory of friction dampers

The scientific study of dry friction has a long history, including the research of Leonardo da Vinci, Amontons, and Coulomb. With the resulting laws, the magnitude of the frictional force can be described as a function of the materials of the two surfaces and the vertical force. This frictional force can be calculated using the following equation:

$$F = \mu N \quad (1)$$

where F and N represent the friction force and normal force, respectively, and μ is the friction coefficient. The value of this coefficient depends on the nature of the surfaces that are in contact with each other. It is slightly dependent on other variables, such as the contact surface and temperature, at most levels. In other words, an object with weight W begins to move when the lateral force P is greater than the frictional force F (Fig. 1). In each case, the frictional force acts tangent to the plane interface and in the opposite direction of motion [31].

Nevertheless, frictional processes are rarely so simple. The Coulomb friction law is only approximately accurate in practice. For example, the coefficient of friction μ at any given moment depends not only on the material of the components but also on the conditions present at the interface between the two surfaces. Experiments have shown that the coefficient of friction depends on factors such as contact pressure, slip speed, temperature, and number of load cycles. Although all these problems exist, the above relationship is the best theory available for friction dampers.

2.1. LSB connections

There are two types of frictional grooved screw connections in concentric bracing frames: linear sliding and rotary sliding connections. Energy is dissipated through these connections without causing structural failure. In these joints, the friction required to create energy loss is provided by the pretensioning force of the screws. The essential factors that affect the efficiency of SBCs are as follows [31]:

1. Keeping the compression between the sliding plates at the

connection constant.

2. Keeping the slip coefficient between the sliding plates constant.
3. Avoiding fracture formation in the connection.

As there are two slip surfaces in grooved screw joints, there will be friction between components when they reach the slip threshold:

$$F = 2T_b N \mu \quad (2)$$

where F is the slip force (N_{slip}), N is the number of bolts, T_b is the tensile force in a bolt, and μ is the friction coefficient. Given Equation (2), according to the above formula, varying the number of bolts and the tensile force of the bolts results in a wide range of slip forces (Fig. 2).

2.2. Introduction of the FBP friction damper

This study introduces a type of friction damper that is resistant and stable against dynamic forces to dissipate a high percentage of input energy and prevent the destruction and damage of structural elements. The friction damper made in the present study is called a *FBP* damper. This new device should exhibit a consistent dynamic behaviour during the loading time, and even in multiple loading cycles, there should be no change in its performance. To achieve this goal, the new friction damper must be designed so that it takes the following two critical factors into account:

1. The contact surfaces must have sufficient slip resistance.
2. Sufficient movement conditions and displacement in the damper must be provided.

If both conditions are provided in a friction damper, then the performance of the damper will be desirable for vibrational energy dissipation. This damper can create desirable properties at the contact surfaces through the selection of the brake pad, which is a very resistant element and is durable against slipping and frictional forces. This makes the *FBP* damper of a moving T-shaped member move in a particular direction. This member's long slots allows it to slide along the desired length. This dissipating device provides the second condition by choosing a suitable place for the damper. This damper is installed in the location of brace connections and story beams, according to Fig. 23. Based on the above description, the story drift occurs where the damper is installed. A special damper has been designed and produced with a friction force by selecting and designing a suitable geometric design for the lining pads. This damper has been shown to perform optimally under cyclic loads in laboratory tests.

3. Experimental programme

This study initially presents a suitable geometric shape for installing an *FBP* damper inside a chevron frame. Then, by performing parametric studies on essential variables, the optimal damper was determined

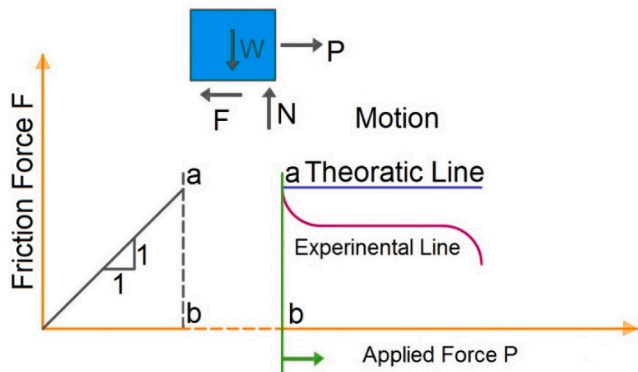


Fig. 1. Relationship of the frictional force with the applied force . adapted from [30]

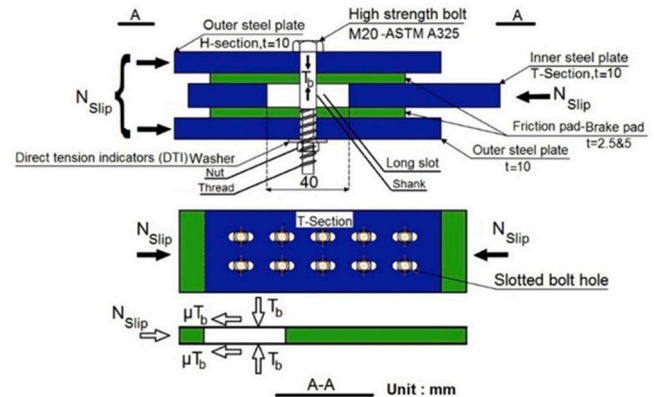


Fig. 2. Details of the SBC (frictional connection with long slotted holes) . adapted from [30]

(Fig. 22), constructed, and tested in the laboratory. In this study, the proposed friction damper consists of two plates of brake pads with steel plates. The pads measure $500 \times 500 \times 5$ mm and are of standard type. The new brake pads are free from asbestos, avoiding health hazards, contamination, and cancer risk. Therefore, these brake pads can be considered organic and environmentally friendly. This pad type also contains nine main elements: steel wool, copper powder, zirconium silicate, vermiculite, phenol formaldehyde resin, aramid fibres, iron powder, silica, baryte, and some composite materials. Due to the production of suitable dimensions, this type of brake pad can be used in the construction industry, especially for friction dampers. The FBP damper, because of its favourable mechanical properties, such as a high friction coefficient, has a high efficiency and high energy absorption in the chevron frame. Additionally, the geometric shape and the proposed combination for the friction damper have helped to optimize and increase the efficiency of this passive control system.

Fig. 3 shows the details of various parts of the friction damper and its different components, including the steel plates, bolts, and brake pad plates.

According to Equation (1), two parameters, the friction coefficient of the slip surfaces μ and the force perpendicular to the N surface, must be calculated to estimate the slip force. On the other hand, the force perpendicular to the surface also depends on the prestressing force of the screws, which is estimated according to the torque required to tighten the screws. In the above damper, by estimating an approximate relationship between the torque required to tighten the screw and the slip force, a simple relation can be obtained to estimate the slip force.

The prestressing force of each screw according to ANSI/AISC 360-16 [32] for M20 bolts of type ASTM A325 is at least 142 kN. Therefore, based on the choice of the 10-screw damper, the total prestressing force was 1420 kN.

The prestressing force in the laboratory model was applied as follows.

Using this method, a twisting moment was applied to the screw with a manual or mechanical torque wrench fitted with a device that measures the twisting moment M_v . Following the initial tightening of the screw, the degree of the torque wrench was set to zero, and the torque wrench was rotated to apply the M_v twisting moment to its nut. The prestressing force was calculated according to the following equation:

$$T_b = M_v / k d_b \quad (3)$$

In this relation, d_b is the diameter of the screw, and k is the

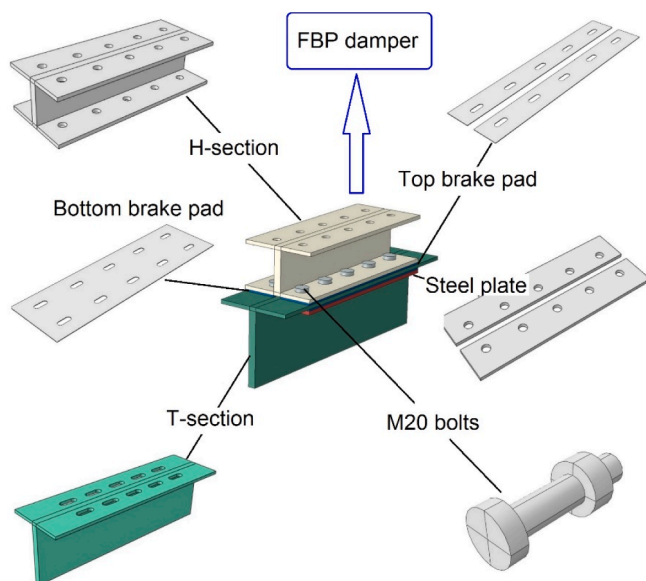


Fig. 3. Damper details in a 3D view.

coefficient for the handle of the torque wrench. Because the accuracy of this method is approximately 10% to 15%, the design regulations recommend that the twisting moment on the screw be adjusted so that the tension created in the screw is 5% higher than the recommended values. Electric and pneumatic wrenches can also be used instead of manual wrenches. Such a torque wrench is applied with the help of electric motors or compressed air. Additionally, considering the use of direct tension indicator (DTI) washers in this test, it was ensured that the screws were prestressed. Therefore, these washers had protrusions that were calibrated for each screw size. Initially, the washer was placed inside the joint and then tightened with the screw; then, by applying more pressure to the prestressed screw, the washer protrusions were eliminated. The complete flatness of the washer is a sign of reaching the minimum amount of prestressing for each screw. One of the simplest methods to apply the prestressing force in ABAQUS software is to select the bolt load in the create load section (Type in a selected step). In this section, the magnitude of 142 kN was applied for each screw.

In this study, two samples of friction dampers were tested under cyclic loading (Table 1).

On the one hand, the motion of the damper and the force of friction between the contact surfaces cause the production of thermal energy at the contact surface of the body. Since it has been repeatedly observed that the coefficient of friction at the slip onset threshold differs somewhat from the coefficient of friction during slip, two coefficients of friction can be introduced, namely, the static friction coefficient μ_s and kinetic friction coefficient μ_k .

As noted earlier, the coefficient of friction depends on many factors and varies during the slip process. In this test, the coefficient of friction assumed in the software is 0.25 based on the material of the selected brake pads, and then, after the test, the average coefficient of friction obtained in both damper models was 0.21. Both specimens had a friction coefficient of 0.21 and differed only in thickness. The difference between the initial assumption and the laboratory result is 16%. This difference arises because of some assumptions about the prestressing force and numerical model, which is common in such experiments.

Table 1 shows the ten-screw damper model with two thicknesses of 2.5 and 5 mm selected for manufacturing and testing in the laboratory. According to Fig. 8, two samples of FBP dampers were tested in the present study under cyclic loading. By following parametric studies, the variables of the current study were examined, and an optimal design for a friction damper was achieved. In this research, friction damper brake pads and a damper with a unique geometric shape and form were used as an innovative part of the research. Good mechanical properties, such as high friction coefficients and increased efficiency and energy absorption, were demonstrated in the bracing frame equipped with dampers. In addition, the geometric shape, along with the dimensions proposed for the friction damper, helped to optimize and increase the efficiency of this passive control system.

According to previous studies, abrasion-resistant materials are recommended for slip-resistant surfaces. Brake pads have been used as intermediate materials to improve friction damper performance. The brake pad and steel plates were studied prior to the test. Additionally, displacement transducers (LVDTs) have been used to calculate displacements. In this experiment, 50 mm and 300 mm LVDTs (SPD-50E and SPD-300E) manufactured by the Tokyo measuring instruments lab

Table 1
Samples of friction dampers.

Model name	Number of brake pad layers	Thickness of brake pads (mm)	Number of screws	Number of steel plates	T_b (kN)
B-10-A-2.5	2	2.5	10	2	142
B-10-A-5	2	5	10	2	142

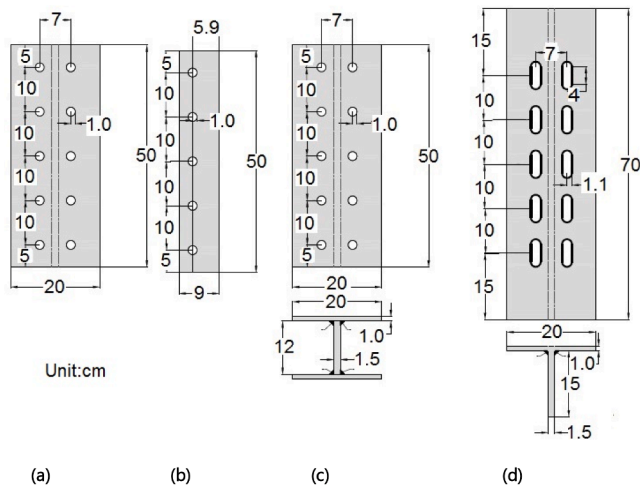


Fig. 4. Damper details **a** brake pad plate at the top, **b** brake pad plate at the bottom and steel plate, **c** H-section (girder) and **d** T-section.

(TML) were utilized. A 50 mm LVDT was installed at the bottom of the *FBP* damper, and a 300 mm LVDT was installed at the top of the *FBP* damper. The SPD-50E model has a sensitivity of 100×10^{-6} strain/mm, and the SPD-300E model has a sensitivity of approximately 33×10^{-6} strain/mm. In addition, a load cell CLC-3MNA from the TML company with a capacity of 3000 kN was used to calculate the amount of applied force. Furthermore, the hydraulic jack used in this test has a load capacity of 800 kN, which can be overloaded by 20% (Figs. 6 and 7).

The properties of the materials were investigated using a STD-1000 universal testing machine with a capacity of 1000 kN. This device's

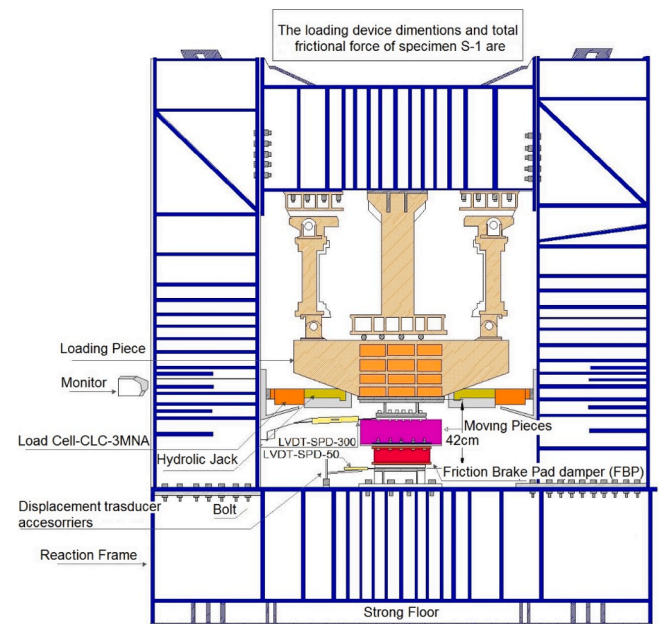


Fig. 6. General overview of the test setup used for the *FBP* damper specimens in the laboratory.

transmission is hydraulic. However, it does not have the disadvantages of (conventional hydraulic) devices with one-way jacks, and its highly advanced control system can guarantee the load rate at very low speeds of 0.01 mm/min (Fig. 9).

ABAQUS simulations of brake pads and metal dampers are not possible without first testing the materials. For this reason, eleven steel

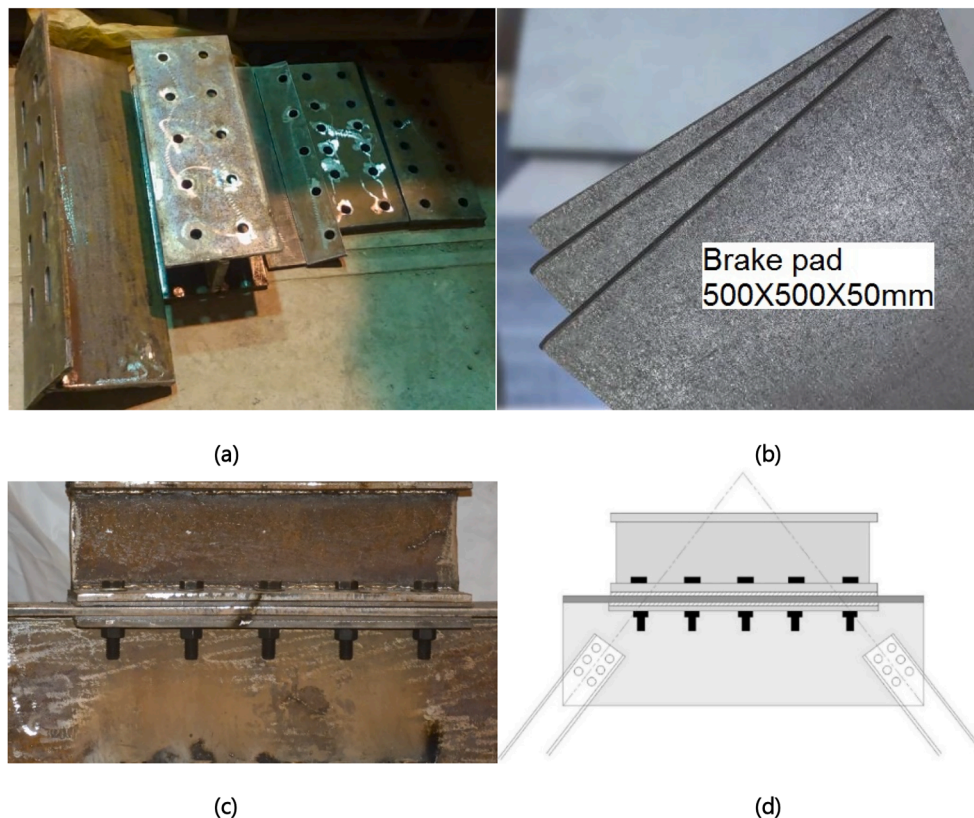


Fig. 5. Laboratory specimen of the friction damper with a brake lining pad **a** laboratory friction damper components, **b** brake pads, **c** built sample and **d** arranged steel plates and brake pads.



Fig. 7. Experimental configuration for the cyclic loading of the friction damper.

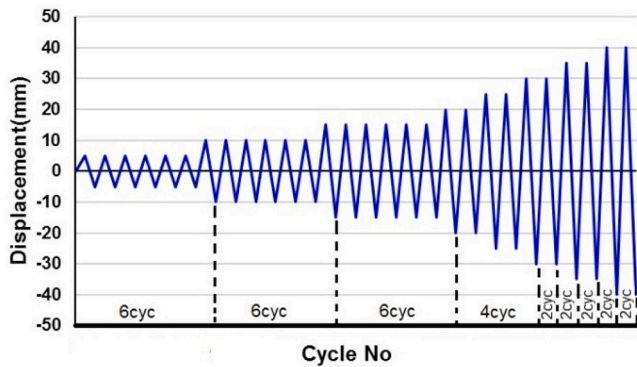


Fig. 8. Diagram of loading protocols.

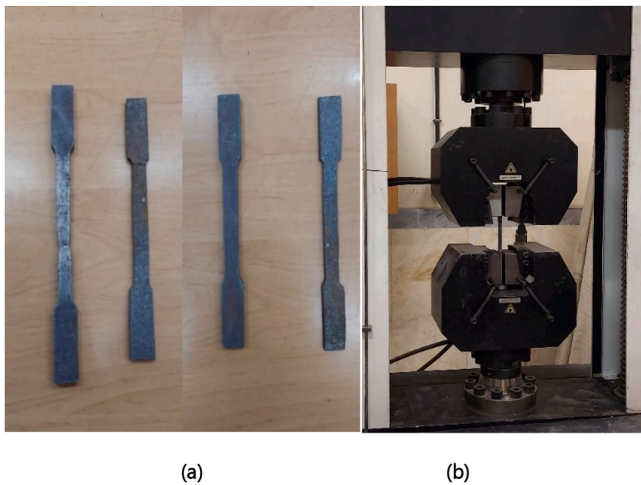


Fig. 9. Coupon test: **a** sample brake pad and steel and **b** universal testing machine.

samples were tested on the universal testing machine (Tables 2 and 3), and the final results of the test are shown in Fig. 10. Metal tension was determined according to the ASTM E8 [33] standard. Based on the results obtained from testing brake pad friction dampers and the software hysteresis diagrams, Fig. 11 shows a minimal percentage difference due to proper meshing (Fig. 12).

Table 2

Mechanical properties of the chevron element.

Elastic modulus (MPa)	Yield stress (MPa)	Ultimate stress (MPa)	Density (kg/m ³)	Poisson's ratio
200	411	488	7850	0.3

Table 3

Coupled test results.

Component	type	F _y (MPa)	F _u (MPa)	F _y /F _u	Elongation (%)
Steel plate	ST37	295	415	0.71	27
Bolt	M20-A325	660	810	0.81	14

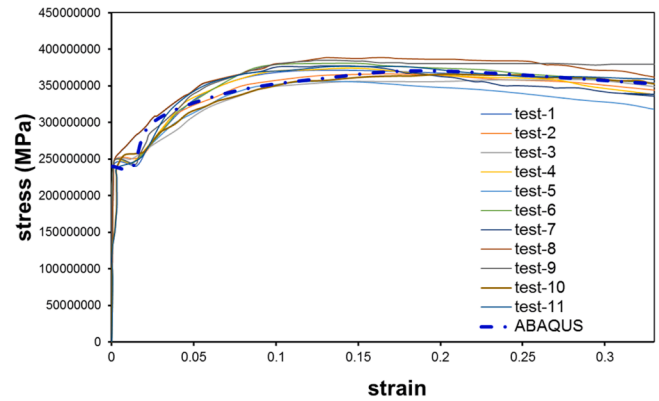


Fig. 10. Strain–stress diagram of steel in the laboratory.

4. Frame verification

In this study, a chevron brace tested by Sen et al. [34] was used to validate the results of the finite element modelling performed in this study. Fig. 13 shows the model's connection geometry. Elements were modelled as solids. The mesh was quite regular. The merge command modelled the connections, assuming they were properly welded. In Fig. 13, V_{bb} stands for story shear at initial brace buckling. Fig. 13 shows the results of the chevron frame validation. Table 4 shows that the difference between the results of Sen et al. [34] and the numerical results is approximately 3 to 15%. This difference may be due to plumbing, material strength, or the approximate solution of differential equations by software. The results in Table 4 show that the model has good accuracy.

5. Parametric study

In this study, the finite element numerical method was used via ABAQUS software. Additionally, since the research variables include the thickness of the brake pads, the arrangement of the plates, the (uniform or nonuniform) distribution of the screws, and the number of screws, nine models were modelled and analysed using ABAQUS. Three different arrangements of screws and two different thicknesses of brake pads were tested; Table 5 shows the specifications of the modelled samples, while Table 6 shows the specifications of the modelled samples with the different numbers of pages and arrangements of plates and screws. Finally, the optimal model was selected. All the components related to the connection, including beams, columns, connection parts, and screws, must be modelled separately when modelling friction dampers with brake pads. Steel beams, brake pad plates, and connection screws were modelled using the C3D8 element. Fig. 4 shows the geometric model of the damper components. Fig. 5 shows the completed damper model, which includes the steel beam, the connecting screws, and the brake pad plates. The names of the models is shown in Table 5. The B symbolizes the first letter of the word brake pad; the next number indicates the number of screws used, and the last number indicates the

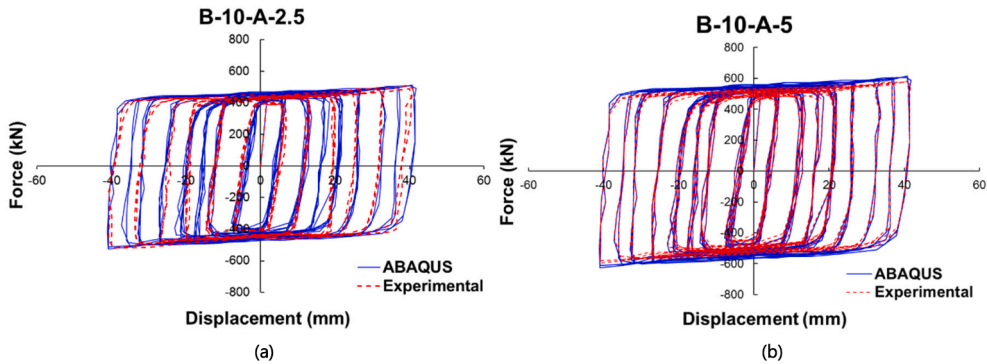


Fig. 11. Hysteresis curve of friction dampers a B-10-A-2.5 and b B-10-A-5.

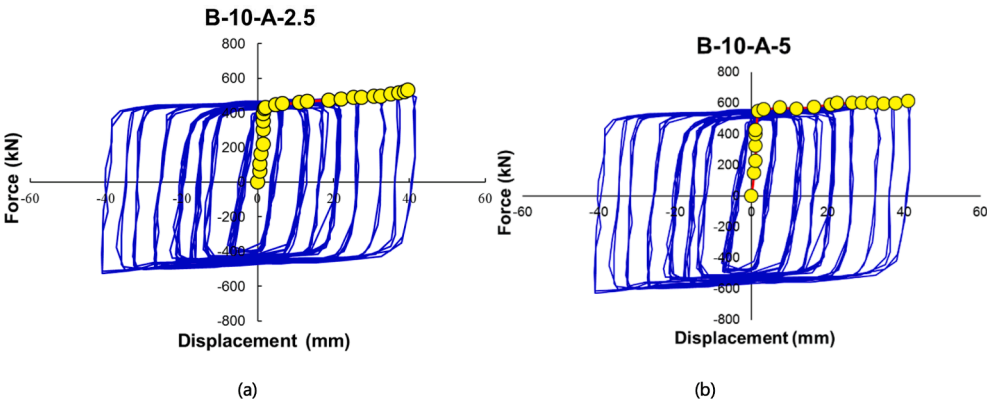


Fig. 12. Backbone curve of friction dampers a B-10-A-2.5 and b B-10-A-5.

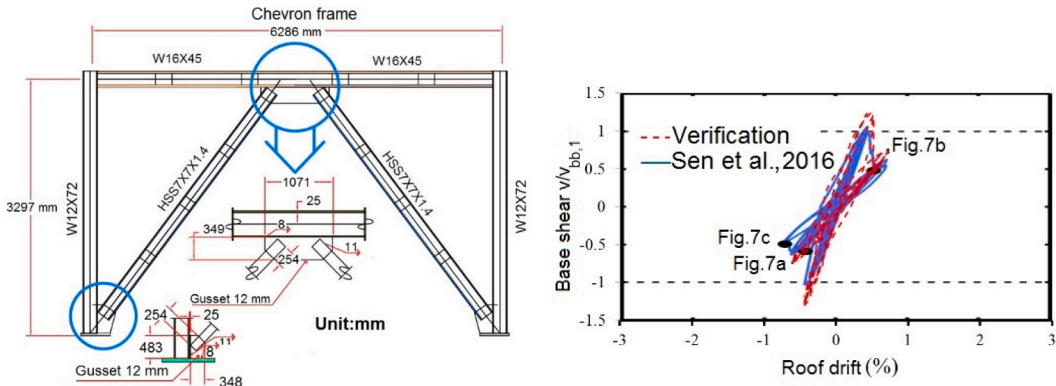


Fig. 13. Base shear roof drift curve of the chevron frame [34].

Table 4
Frame parameter verification.

Difference between ABAQUS and Sen et al. [34] (%)			
v/v_{bb}	Area enclosed by a hysteresis	Roof drift	Stiffness
12	15	3.32	2.56

thickness of the brake pad used. Model B-6-A-2.5, for example, means a brake pad damper with six long slot holes and a 2.5 mm asbestos thickness. The reason for selecting this thickness is that brake pad manufacturers are more likely to produce and offer the two thicknesses

Table 5
Specifications of the modelled samples to determine the optimal model.

Model name	Number of screws	Thickness of brake pads (mm)
B-6-A-2.5	6	2.5
B-10-A-2.5	10	2.5
B-14-A-2.5	14	2.5
B-6-A-5	6	5
B-10-A-5	10	5
B-14-A-5	14	5

Table 6

Specifications of the modelled samples with different numbers of sheets and arrangements of sheets and screws.

Model name	Number of brake pad layers	Thickness of brake pads (mm)	Number of screws	Number of steel plates
B-10-A2.5-3	3	2.5	10	2
B-10-A2.5-4	4	2.5	10	2
B-10-A2.5-2	2	2.5	10	1

of 2.5 and 5 mm; thus, this choice reduces the cost of brake pads. Notably, brake pad thicknesses less than 2.5 mm have a lower shear strength coefficient, and brake pad thicknesses greater than 5 mm have little impact on improving frictional conditions or increasing the efficiency of this type of damper.

5.1. Meshing and its optimization

Meshing the various damper elements was one of the first steps in the finite element model analysis. Finding the optimal mesh size plays an essential role in these analyses. To mesh the damper model, first, an initial meshing of the elements was performed. We analysed the damper model under the desired load and extracted a specific point's stress and displacement in the damper model. Following this, the mesh size was reduced, the model was reanalyzed, and the results were recorded. Finally, the convergence curve was drawn using the results of the stress and displacement measured while considering changes in the mesh size. Because of the friction of the surfaces, *C3D8R* meshing was selected. A finer mesh was used to obtain suitable results from the stress analysis at the model's connections. For the meshing of steel beams, metal plates, and brake pads, the *C3D8* eight-node element was used (Fig. 14). The *C3D8* type was not possible in some areas because of the particular geometry used; in these cases, the *C3D10* ten-node element was used. Fig. 15 shows an optimal mesh diagram.

5.2. Loading protocol used

$$\mu_{max} = \frac{\Delta_u}{\Delta_y} \quad (4)$$

Before selecting the appropriate loading protocol, the first nonlinear static analysis was performed by modelling the pushover load for friction dampers in *ABAQUS*. Then, the yield strength and the final limit displacement were calculated according to Fig. 16. Next, then based on Equation (3), a ductility equivalent to 6.37 was calculated. Accordingly, the appropriate loading protocol was selected and applied. AISC-341-10 or SAC loading protocols have been used as standard loading protocols in various tests for many years. The advantage of the SAC loading protocol is that it is more commonly used to test dampers and steel structures because of its more straightforward loading pattern. In the SAC protocol, because the loading speed and round-trip cycles are initially much slower than in other protocols, better energy absorption is

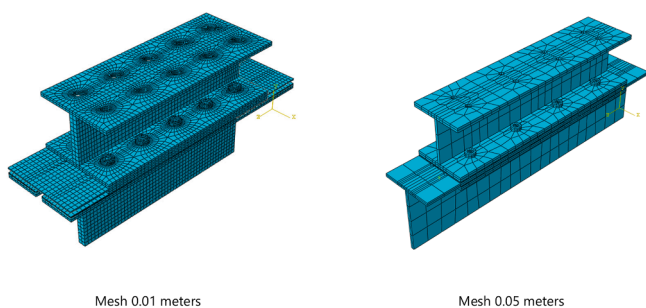


Fig. 14. Complete damper mesh with total connections in ABAQUS.

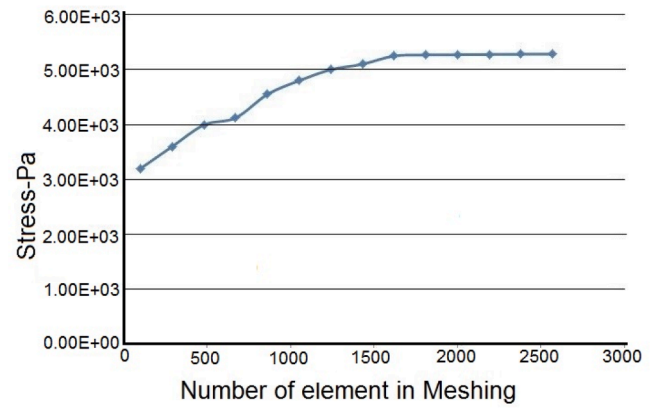


Fig. 15. Optimal meshing diagram.

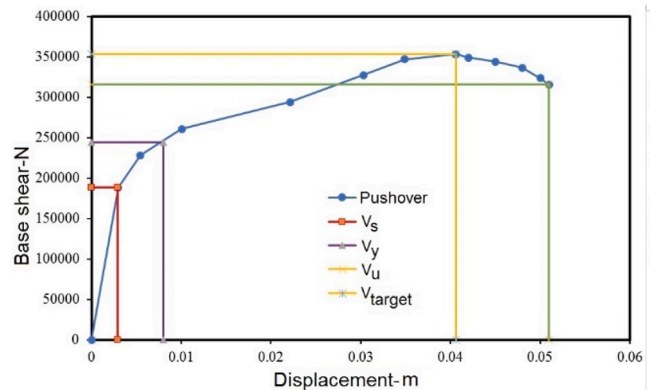


Fig. 16. Pushover curve of the brake pad damper.

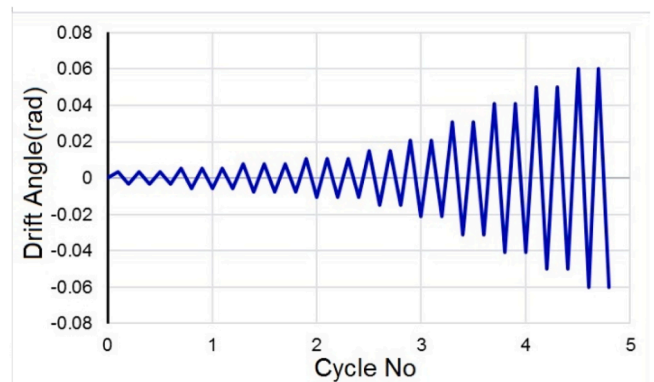


Fig. 17. Loading cycles on samples according to the AISC-341-10 [35] protocol.

observed in the damper.

The SAC loading protocol has been used in two ways, namely, rotation and displacement, and the criterion utilized in this study was the displacement. In this protocol, the rotation criterion was used to connect the beam to the column. The control parameter was the story drift angle in the SAC loading protocol. The loading range started with six deformation cycles with drifts of 0.35, 0.5, and 0.75% and continued with four cycles with a 1% deformation range. In the next step, two cycles with a 1.5% deformation range were created, and in the next steps, two cycles with deformation ranges of 2, 3, 4%, etc., were created. One of the essential features of the SAC loading protocol is a proper simulation of earthquake effects on steel joints and dampers. Fig. 17 shows the SAC loading protocol used in damper analysis.

5.3. Results of damper model analysis

As shown in Figs. 18 and 19, *von Mises stress*, one of the most significant indexes in discussing finite element analysis results, occurs in the damper. According to the results, the lower and upper brake pads had higher stress concentrations. Fig. 18 shows the result of the elastic strain in the brake pads, showing that the strain distribution was very uniform.

With small displacements, such as those caused by weak earthquakes, the *FBP* damper can provide fast energy absorption. Its energy absorption increases with displacement, i.e., in severe earthquakes, it can absorb much more energy. For example, Fig. 20 shows a 22% increase in force at the ultimate displacement for the B-10-A-2.5 damper (Fig. 21). The friction damper has demonstrated good performance under small and large displacements by absorbing the energy caused by the earthquake by friction, which suggests that this type of damper can

be used for any earthquake magnitude. As a result of analysing the friction damper without the frame in the finite element software, the load is applied directly to the web of the T-section. The other elements remain motionless, and only the T-section between the upper and lower brake pads and the steel sheet move and create friction. Hence, the T-shaped piece experiences an equivalent plastic strain. According to Fig. 19, friction dampers initially create stress concentrations as a result of the movement mechanism of the T-section along the length of long slots and around the connection. This was a predictable outcome. The first section to fail after the bolts, for example, when there is a substantial displacement or force applied, is the T-section. Due to the construction and installation of the friction damper and the fact that all connections of this set are bolted, the damper can be moved, allowing the damaged components to be replaced after an earthquake. Under a low force or a small displacement, the frame absorbs most of the force. However, when the displacement of the frame increases due to an

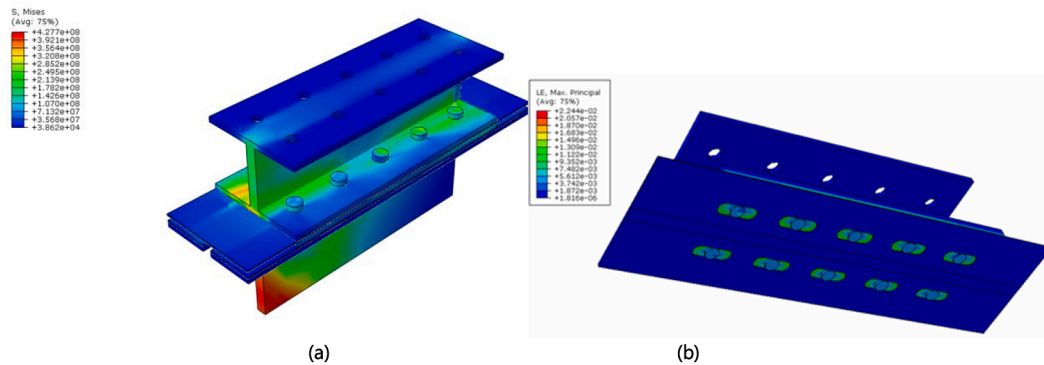


Fig. 18. The finite element output an elastic strain **a** on the top brake pad, **b** von Mises stress in the FBP damper.

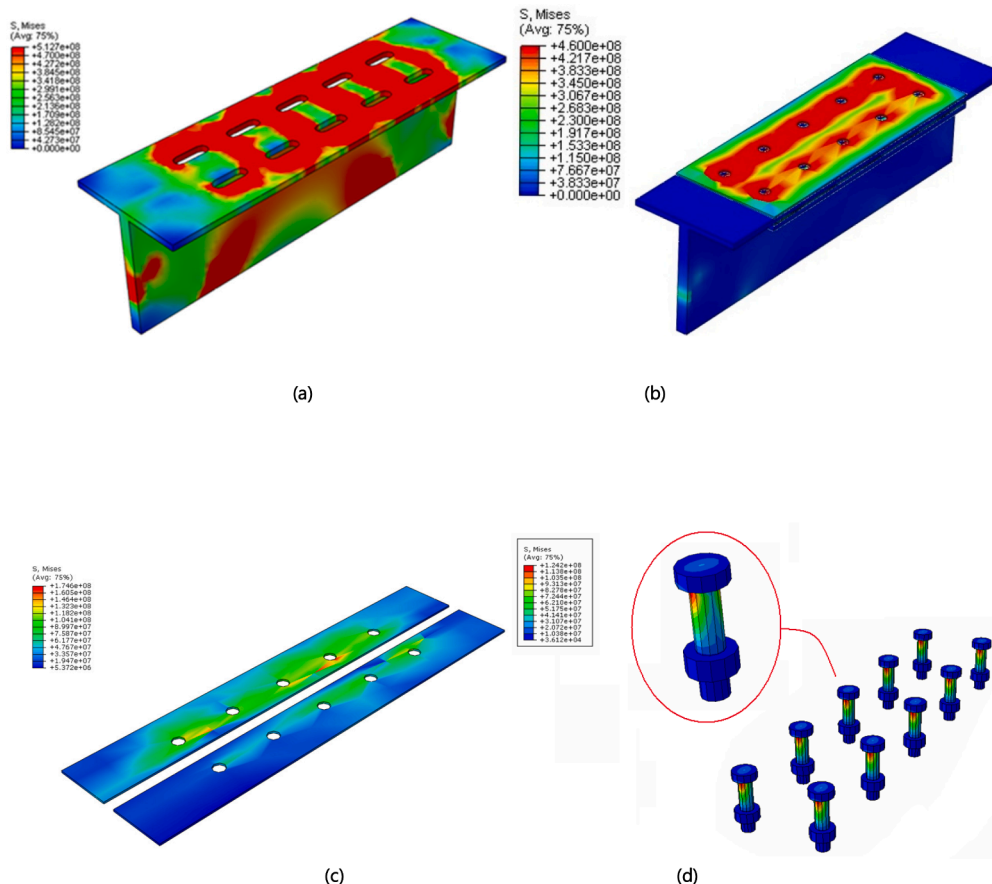


Fig. 19. Von Mises stress in **a** the top brake pad, **b** the T-section, and **c** the bottom brake pad and **d** stress distribution in the bolts in the model.

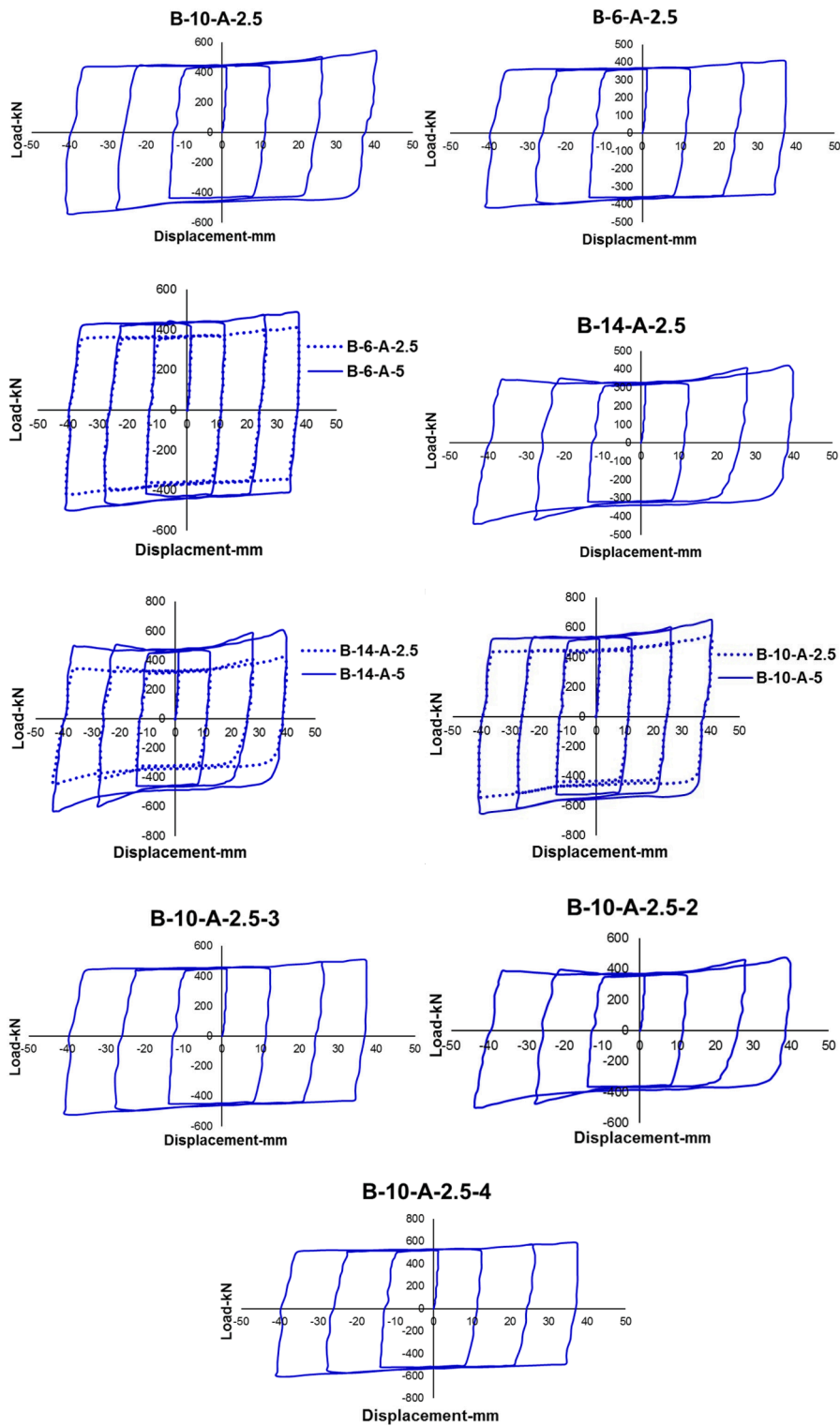


Fig. 20. Hysteresis curves of the FBP damper.

earthquake and the displacement exceeds the elastic limit of the damper, damage occurs in the damper; previously, the displacement mechanism was linear, and the damper was elastic. When a weak or moderate earthquake occurs, the frame, the damper, and their entire set will not fail.

5.4. Output of hysteresis curves

Fig. 21.

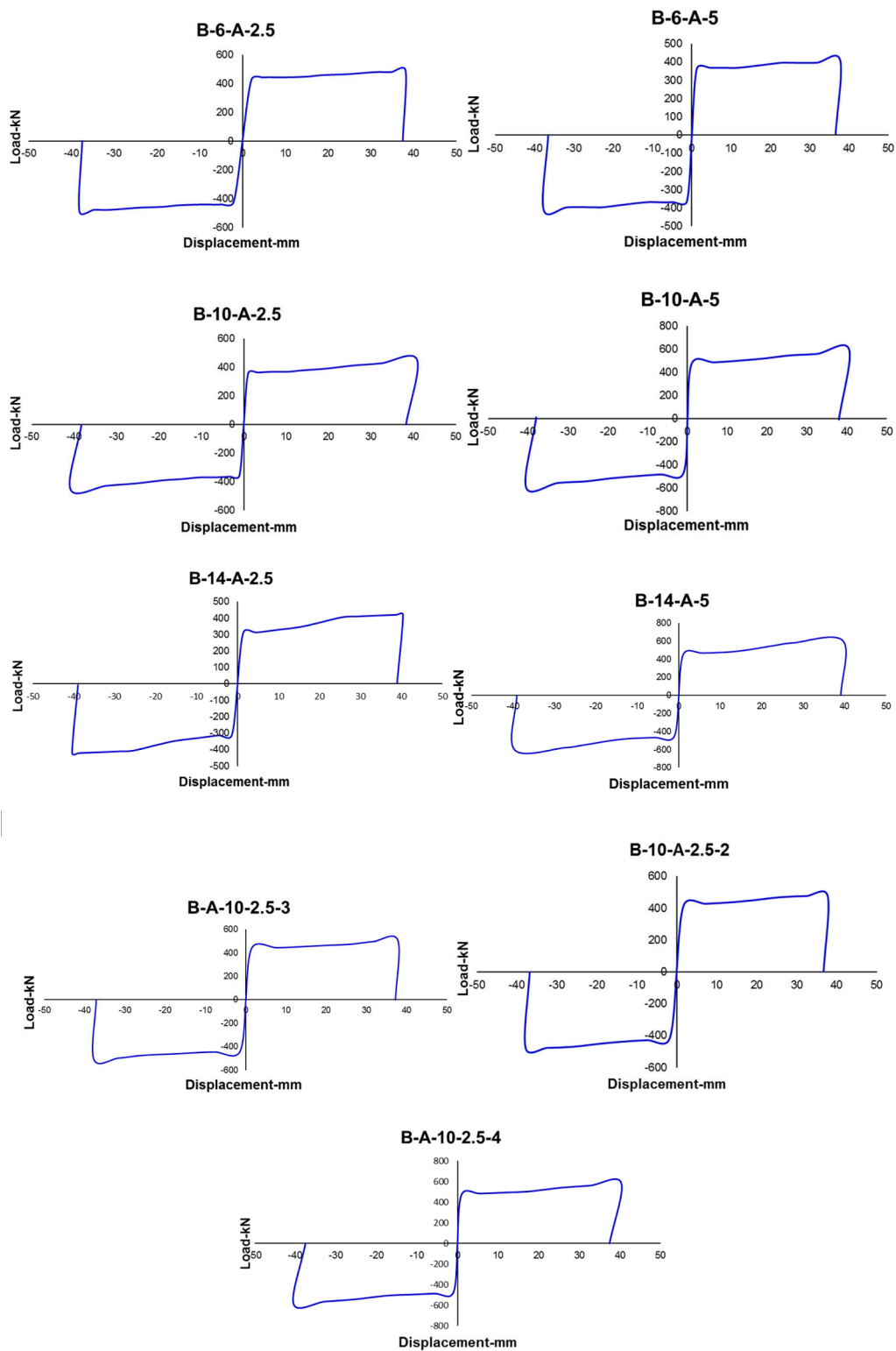


Fig. 21. Backbone curves of the FBP damper.

5.5. Backbone curves

Fig. 22.

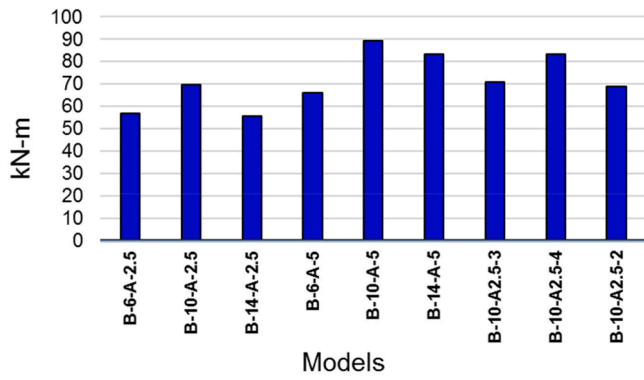


Fig. 22. The level of energy dissipated by each damper.

6. Friction damper performance of a chevron frame.

In this study, a chevron brace tested by Sen et al. [34] was used to estimate the behaviour of a chevron frame equipped with an *FBP* damper. Fig. 23a shows the model's details.

The semirigid beam-to-column connections were selected to better examine the effect of the *FBP* damper on the chevron frame. Thus, the frame alone contributed little to the lateral resistance, and the results show the *FBP* damper's net performance.

According to Fig. 23b, showing the von Mises stress created in the model, the maximum amount of stress occurs at the connection between the damper and the beam ($4.8 \text{ e}8 \text{ N/m}^2$). The amount of stress at the point of connection of the beam to the column is $2.683 \text{ e}8 \text{ N/m}^2$ due to the use of the *FBP* damper. The chevron frame without a damper shows that the maximum stress occurs at the connection between the beam and the column, and this value is equal to $4.8 \text{ e}8 \text{ N/m}^2$. In other words, the presence of a damper reduces the stress in the area of the beam-column connection by approximately 40%. As a result of installing the *FBP*

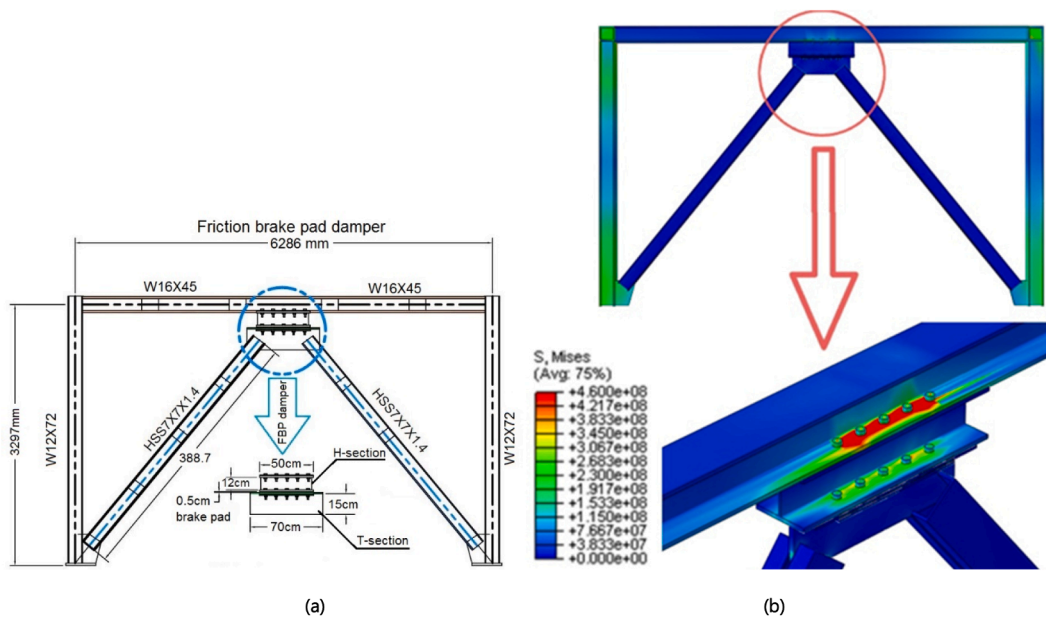


Fig. 23. FBP damper in chevron frame and b von Mises stress in the chevron frame.

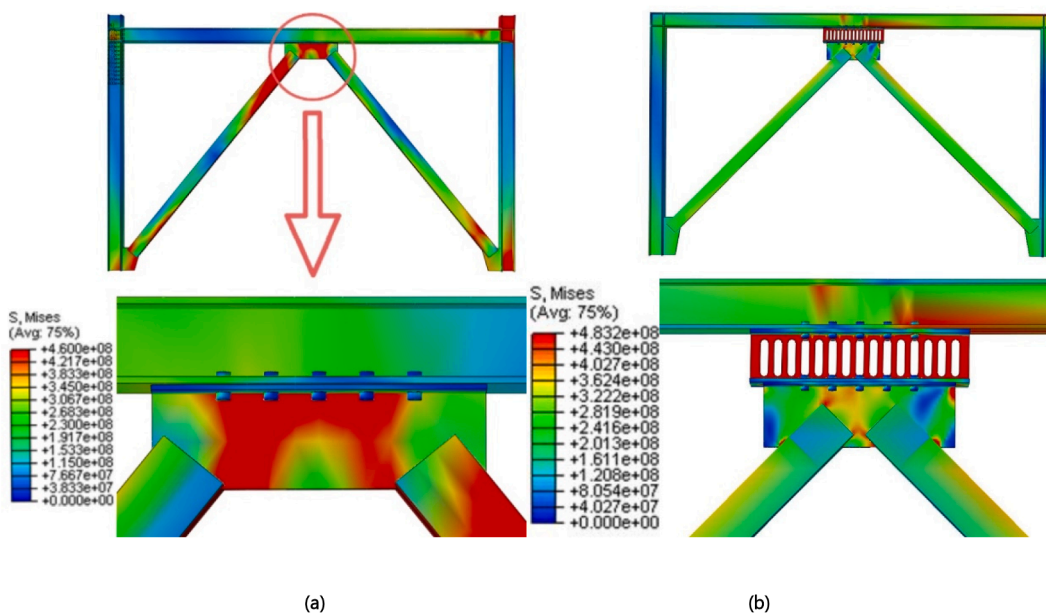


Fig. 24. Von Mises stress in in the chevron frame a without a damper and b with an ADAS damper.

damper in this frame, the amount of von Mises stress in the connection area decreases, and this stress is created in the *FBP* damper area. Other elements, such as beams and columns, are less stressed because dampers are present. The applied force in the frame is converted to a lateral force in the damper, which causes the damper to move (in light of the distances and slotted bolt hole created in the damper system), causing friction between the inserted pads (Fig. 24).

According to Table 7, the chevron frame with the *FBP* damper has a higher elastic stiffness, overstrength, and response modification factor than the chevron frames with *ADAS* dampers and without dampers. This difference is small for one-story frames, and with an increasing number of stories, the efficiency of this system increases. For example, ductility coefficients in structures depend on the ultimate and yield displacements. The *FBP* damper, according to the nonlinear time history analysis, has been shown to be more effective than other passive control systems at higher altitudes with greater displacements. The dissipated energy sector in Fig. 25 and the base shear in Table 7 are significantly different with regard to passive controls. The passive friction damper control efficiency of conventional buildings will be enhanced due to these structural parameters. Fig. 23 shows that the distribution of von Mises stresses in the connections does not cause damage, the damper reduces the resulting stresses, and the effect of the *FBP* damper is observed in the frame.

Table 7

The structural parameters in the cases of the frame without a damper and frame with a damper.

Seismic parameters, meaning	Seismic coefficients	Chevron frame without damper	Chevron frame with <i>ADAS</i> damper	Chevron frame with <i>FBP</i> damper	Units
	Parameters				
Base shear related to the first plastic hinge formation	V_s	410	490	500	kN
Displacement related to V_s	Δ_s	0.02	0.02	0.02	m
Initial stiffness of push-over curve	K_E	20,500	24,500	25,000	kN/m
Maximum base shear	V_0	510	683	726	kN
Displacement related to V_0	Δ_m	0.15	0.15	0.15	m
75% maximum base shear	$0.75 V_0$	382.5	512.25	544.5	kN/m
Displacement related to 75% V_0	Δ_1	0.020	0.023	0.028	m
Displacement related to yielding force	Δ_y	0.027	0.037	0.030	m
Elastic stiffness	K_0	13,721	14343.7	17755.4	kN/m
Surface below the bilinear diagram	A_2	69.7	89.7	97.7	m ²
Displacement related to V_E	Δ_E	0.084	0.085	0.088	m
A structure's base shear force during elastic behaviour	V_E	1649.7	2096.5	2210.9	kN
Overstrength coefficient	R_s	1.24	1.39	1.45	—
Ductility coefficient	R_μ	3.23	3.04	3.07	—
Response modification factor	R	4.02	4.23	4.46	—
Ductility	μ	5.55	4.05	5.00	—
Area enclosed by a hysteresis loop	Energy	232.94	263.28	308.63	kN.m

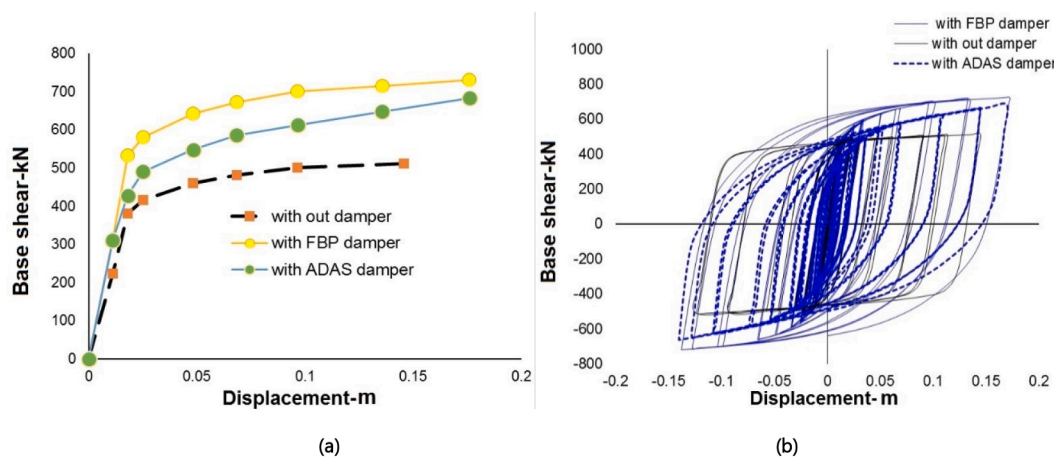


Fig. 25. Comparison of the curves of all three frames: **a** hysteresis curves and **b** backbone curves.

6.1. Compare curves

Fig. 26. Fig. 27. Fig. 28. Fig. 29. Fig. 30.

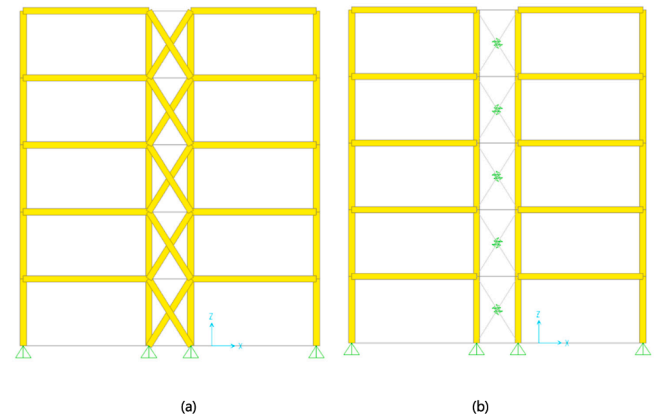


Fig. 26. Structural model of two adjacent MDOF buildings connected with **a** CBF and **b** an FBP damper.

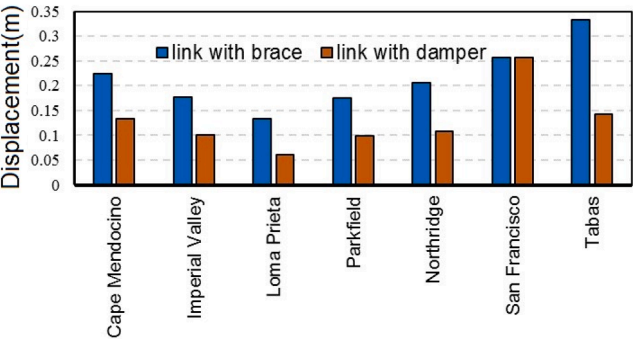


Fig. 27. Comparison of the displacement of the two MDOF buildings.

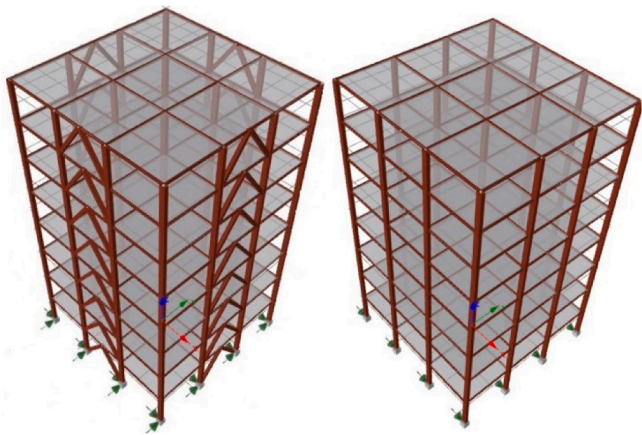


Fig. 29. 3D view of the 8-story structure in the MRF system and chevron bracing with the FBP damper.

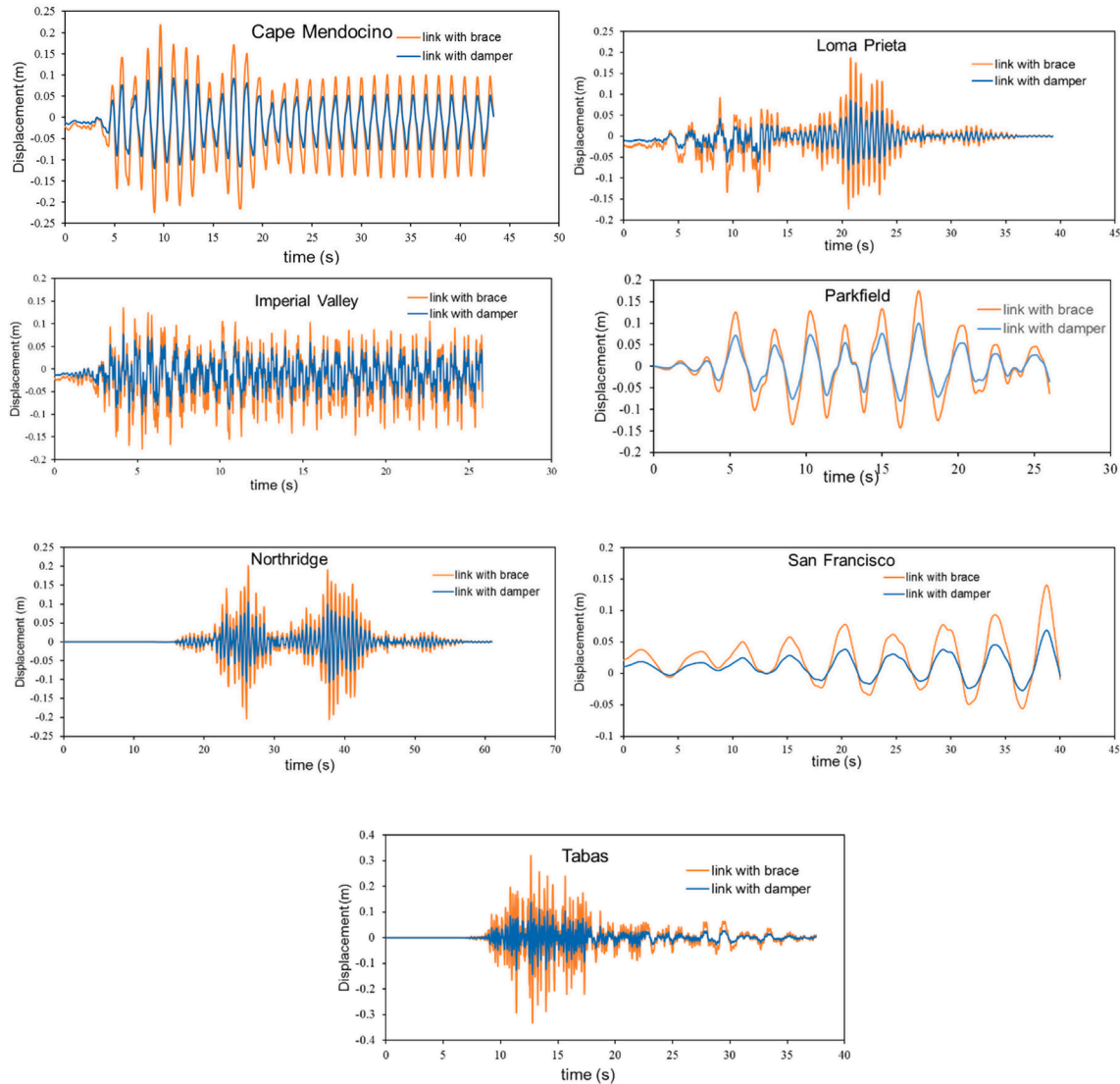


Fig. 28. Time histories of the top floor displacements of the two MDOF buildings.

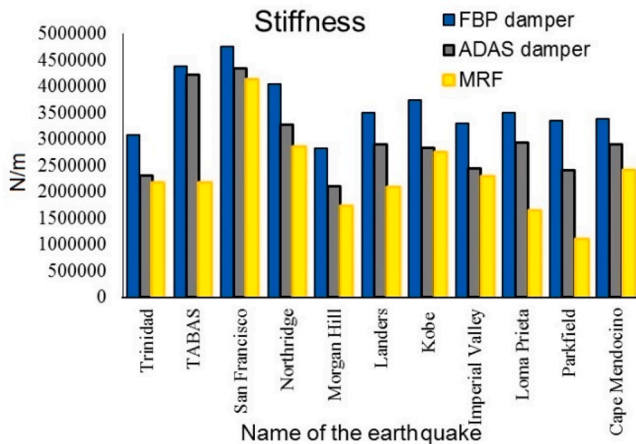


Fig. 30. Stiffness of 3 structures for eleven earthquake records.

6.2. The effect of time and heat on damper performance

Based on studies conducted from reliable sources, the effect of time on the properties of materials used in dampers is of great importance. The time factor alone does not affect the material's properties and therefore does not affect its performance. The reports are based on the effect of heat on the properties of materials and the duration of heat application on their mechanical properties. Usually, this only happens when the building catches fire near the damper installation site. The damping performance can then be evaluated by altering the material's stress-strain curve and reanalyzing the finite element model. However, the damper retains its original performance over time.

7. Seismic response of adjacent buildings connected with FBP dampers

The seismic response of friction dampers associated with adjacent buildings was studied by Bhaskararao and Jangid [11]. Previous analyses and experimental research have already shown the effectiveness of passive energy dissipation systems for improving the seismic performance of connected buildings. However, the performance of adjacent buildings with FBP dampers has not been studied. Therefore, in this section, the structural responses of two adjacent buildings connected in one place with FBP dampers and with cross-bracing frames (CBFs) were analysed using *OpenSees* software under seven far-field earthquake records. Then, the results were compared.

To investigate FBP damper behaviour, two 5-story frames with 6-m spans and 3.2-m heights were modelled and studied at a distance of 2 m from each other (Fig. 26). The results of the time history analysis with the selected earthquake records are as follows:

According to the results, using FBP dampers to connect adjacent buildings with different frequencies can effectively reduce their seismic responses. It is also unnecessary to connect adjacent buildings on all floors, but fewer dampers in suitable locations will significantly reduce the combined system's seismic response.

8. Nonlinear time history analysis

This section discusses the optimal damper model for a middle-class 8-story structure with a chevron bracing system based on parametric studies and laboratory testing. The effects of FBP dampers are also examined. X- and Y-direction earthquake records were applied to the buildings. All records were taken from the Pacific Earthquake

Engineering Research Center (PEER) database. Eleven earthquake records were selected. The earthquakes had Richter magnitudes between 5.28 and 7.35. Table 8 summarizes the earthquake records used.

In addition, this type of FBP damper system in the middle of the structure was compared with other structural systems. To achieve this, the first three structural systems were analysed and designed, including the chevron bracing system with the FBP damper, the MRF structural system, and the bracing structure with the ADAS damper (Figs. 30, 32, 33). In Table 8, the results of the nonlinear time history analysis that was performed on eleven records of far-field earthquakes are presented. In terms of displacement, structures equipped with FBP and ADAS dampers exhibit a much lower displacement than that of the MRF structures. The friction dampers of brake pads have a very high ability to dissipate seismic energy, as shown in Fig. 31.

Table 8

Records used to analyse time history.

ID No.	Earthquake			Station	Soil Data	Site-Source Distance
	Magnitude	Year	Name	Name	NEHRP	Joyner Boore
1	7.2	1980	Trinidad, Caribbean	Rio Dell Overpass	311.75	76.06
2	7.35	1978	Tabas, Iran	Bajestan	377.56	120.81
3	5.28	1957	San Francisco, United States	Golden Gate Park	386.25	17.35
4	6.69	1994	Northridge, United States	Alhambra - Fremont School	549.75	36.77
5	6.19	1984	Morgan Hill, United States	APEEL 1E - Hayward	354.2	51.68
6	7.28	1992	Landers, United States	Amboy	382.93	69.21
7	6.9	1995	Kobe, Japan	Abeno	256	24.85
8	7.6	1938	Imperial Valley, United States	El Centro Array	213.44	34.98
9	6.93	1989	Loma Prieta, United States	APEEL 10 - Skyline	391.91	41.71
10	6.19	1966	Parkfield, United States	San Luis Obispo	493.5	63.34
11	7.01	1992	Cape Mendocino, United States	Fortuna - Fortuna Blvd	457.06	19.95

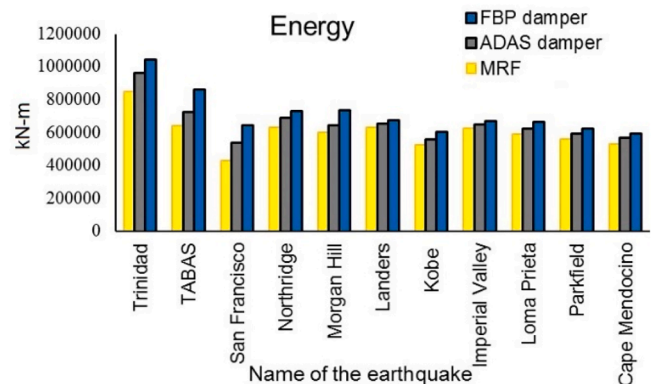


Fig. 31. Energy of 3 structures for eleven earthquake records.

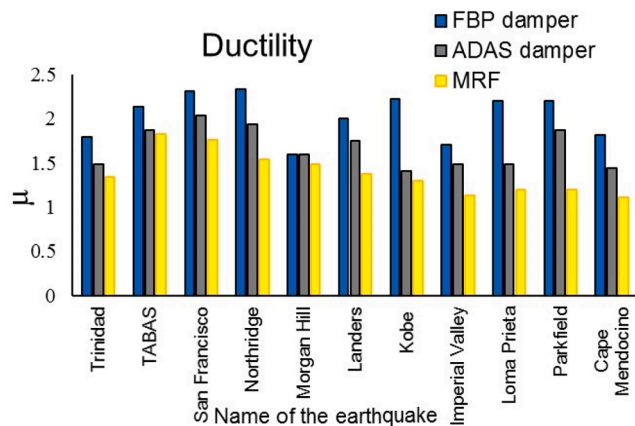


Fig. 32. Ductility of 3 structures in eleven records.

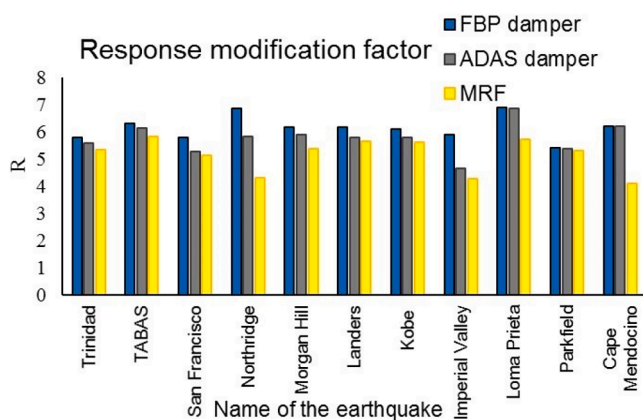


Fig. 33. Response modification factor of 3 structures in eleven records.

9. Conclusion

The friction damper is one of the most effective tools for dissipating the energy of seismic vibrations in structures. In this research, we developed an *FBP* damper element that acts independently of other structural components and has a stable hysteresis behaviour under tension and pressure. This study involved a numerical study of friction damping brake pads. For this purpose, *ABAQUS* software was used to validate a laboratory model. A model was later developed based on this validation work and the use of a brake pad damper. The following results are based on the nine models examined in *ABAQUS* software and two models fabricated and analysed in the laboratory:

1. B-6-A-5, B-10-A-5, and B-14-A-5 are the same as B-6-A-2.5, B-10-A-2.5, and B-14-A-2.5; the only difference is that the thickness of their asbestos layer was doubled, i.e., from 2.5 mm to 5 mm; this increase in thickness improved the energy absorption capacity. However, the increase in energy absorption was not significant. In this case, it seems that thickening the asbestos did not significantly improve damper performance.
2. Based on the frame analysis, the friction damper area of the brake pad experienced the most stress.
3. The model with ten screws consumed the most energy and exhibited the highest energy absorption among the models.
4. According to the analysis, increasing the number of sheets because of a particular mechanism does not increase energy absorption.
5. Adding friction dampers to the frame increased energy absorption. It reduced the force concentration in the braces, which prevents pressure from buckling the friction damper, increasing the chevron frame's performance by 32%.

6. The seismic parameters, including the elastic stiffness, over-strength, and response modification factor, were significantly improved in the chevron frame equipped with an *FBP* damper.

7. A comparison of the energy dissipated in the 8-story structure equipped with the friction damper in the brake pad and in the other two structures shows that the energy dissipated was significantly higher with the friction damper than in the other two structures, demonstrating the efficiency of the passive control system compared to other structural systems.

8. A comparison of the stiffness results of the three structures under eleven far-field earthquake records showed that the bracing system with the friction damper resulted in a relatively high stiffness.

9. According to the laboratory results and the software modelling, the shape of the hysteresis diagram showed no pinching, and the resistance does not deteriorate, which is why this damper is efficient.

10. Brake pad friction dampers are easily replaceable after an earthquake, since they act like fuses.

11. When *FBP* dampers connect adjacent buildings with different frequencies, the direct earthquake response in each structure is effectively reduced.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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